Morphologically Grounded ECG Arrhythmia Classification: A Progressive CNN Pipeline with Physiological Alignment Supervision

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GitHub: https://github.com/hassan7-dev/ecg-morphological-grounding-dl-project.git

Abstract

Deep ECG classifiers routinely achieve high benchmark accuracy by latching on to amplitude statistics and recording-level shortcuts rather than the localised wave morphology that cardiologists actually inspect. This paper works through the problem in three stages on PTB-XL, restricted to Lead II and five diagnostic superclasses. We first establish a 1D-ResNet baseline with Butterworth-filtered inputs and measure saliency quality through a Morphological Alignment Score (MAS) computed against NeuroKit2-derived P/QRS/T boundaries. The first improvement runs a five-condition preprocessing ablation: we designed Lead-wise Beat-Warp Augmentation (LBWA), a beat-preserving augmentation grounded in both the STAR framework and piecewise time-warping theory, but the ablation showed that raw ECG with z-score normalisation alone outperforms every filtered or augmented variant on Macro-AUC, Macro-F1, and MAS. Building on that finding, the second improvement keeps the raw-signal pipeline and adds class-conditioned physiological alignment supervision, which regularises the model's class activation maps toward Gaussian priors encoding the diagnostically relevant wave regions for each arrhythmia. The Full alignment model improves class-specific MAS by 58% relative to the raw-signal baseline with a marginal gain in Macro-AUC (0.8960 $\rightarrow$ 0.8968), and the Conduction Disturbance class, whose defining feature is QRS widening, sees a tenfold increase in region-specific MAS.

1. Introduction

The 12-lead electrocardiogram remains the default first test in cardiac workups, and for good reason: arrhythmias leave specific, readable fingerprints on the waveform. Wolff-Parkinson-White produces a delta wave before the QRS complex. Long QT syndrome extends the QT interval beyond 480 ms. Atrial fibrillation replaces organised P-waves with irregular RR intervals. A cardiologist works through these cues sequentially: inspect the P-wave, measure the PR interval, examine QRS width, then read the ST segment and T-wave.

Deep networks, by contrast, appear to skip this entirely. Ansari et al. (Ansari et al., 2023) surveyed 368 deep learning studies from 2017 to 2023 and found that benchmark Macro-F1 scores are routinely driven by spectral envelopes and amplitude statistics, not the specific morphological intervals that carry diagnostic meaning. Three failure modes compound one another: shortcut learning (majority-class texture dominates gradient attribution), augmentation-morphology conflict (global time-warping distorts PR intervals and QT durations as if stretching artefact mimics pathology), and physiologically naive denoising (Butterworth bandpass filters attenuate the low-frequency P-wave envelope, degrading exactly the feature needed for P-wave-dependent diagnoses).

Our work does not try to match the full complexity of a cardiologist's reasoning process. The narrower aim is to ask whether we can at least align the model's spatial attention with the ECG regions a clinician would look at to justify a given prediction. We quantify this through a Morphological Alignment Score applied at two levels of granularity across three progressive stages.

The specific contributions are:

- A five-condition preprocessing ablation that includes an honest negative result: LBWA, a beat-preserving augmentation we designed by combining the STAR principle with piecewise time-warping theory (Iwana & Uchida, 2021), consistently hurt both classification accuracy and morphological alignment compared to the no-augmentation baseline, while raw ECG with z-score normalisation won across every metric.
- A class-conditioned physiological alignment supervision framework that regularises class activation maps toward Gaussian priors encoding the diagnostically relevant wave regions per arrhythmia class, improving

class-specific MAS by 58% with minimal classification cost.

- An empirical finding that Butterworth filtering, despite being standard practice, measurably suppresses morphological learning compared to processing the raw signal.

2. Related Work

Surveys and the morphology gap. The most complete picture of where the field stands comes from Ansari et al. (Ansari et al., 2023), whose review of 368 studies establishes that 1D CNNs dominate (59% of architectures) and only 28% of studies apply any augmentation at all, let alone morphology-aware augmentation. Crucially, none evaluated whether their augmentation preserved the P-QRS-T ordering that clinical diagnosis depends on.

Explainability baselines. Talukder et al. (Talukder et al., 2025) provide the most directly useful calibration point: their Grad-CAM maps localise to the QRS complex for ventricular arrhythmias and to the P-wave region for supraventricular classes. We adopt this localisation pattern as the qualitative target for our MAS metric. Pantelidis et al. (Pantelidis et al., 2025) release open-source Grad-CAM code and pre-trained PTB-XL weights for a multi-scale Inception-1D baseline, which we use as a post-hoc XAI reference. The practical upper bound on intrinsic explainability is set by ProtoECGNet (Sethi et al., 2025), a prototype-based architecture that justifies predictions via similarity to learned beat exemplars.

Architecture for morphology. Hammer et al. (Hammer et al., 2024) separate a short morphology branch (0.6 s windows) from a long rhythm branch (10 s windows) in their xECGArch, validating separation via pixel-flipping. That morphology-rhythm distinction informs the design of our physiological priors, even though we pursue it through a loss-function approach rather than a dual-branch architecture.

Denoising. The Stationary Wavelet Transform with a Daubechies-8 wavelet is theoretically superior to Butterworth bandpass for preserving P-wave amplitude, because it operates on frequency bands in a shift-invariant way without the subsampling artefacts of the discrete wavelet transform (Addison, 2005; Lee et al., 2019). We tested this empirically and the results are in Section 3.4.

Augmentation. Nemati (Nemati, 2025) introduced STAR, which applies sinusoidal time-amplitude resampling within each RR interval to simulate heart rate variability without distorting intra-beat morphology. STAR was not peer-reviewed at the time of writing, so we grounded our LBWA design additionally in Iwana and Uchida (Iwana & Uchida, 2021), whose empirical study on 128 time-series datasets confirmed that piecewise warping preserves class-discriminative structure better than global warping.

3. Methodology

3.1. Dataset and Evaluation Protocol

All experiments use PTB-XL (Strodthoff et al., 2021), a dataset of 21,799 10-second 12-lead recordings at 500 Hz. We restrict to Lead II (index 1) and the five diagnostic superclasses: NORM, MI, STTC, CD, and HYP. Table 1 shows the class distribution.

Table 1. PTB-XL diagnostic superclass distribution (filtered dataset).

Class	Records	% of set
NORM (Normal)	9,528	44.5
MI (Myocardial Infarction)	5,486	25.6
STTC (ST/T Change)	5,123	23.9
CD (Conduction Disturbance)	4,907	22.9
HYP (Hypertrophy)	2,655	12.4
Train/Val/Test splits: 17100 / 2155 / 2162		

The canonical PTB-XL inter-patient split assigns folds 1–8 to training, fold 9 to validation, and fold 10 to test. Patient-ID intersection between training and test was verified to be zero before each run. Raw accuracy is not reported; NORM exceeds 44% of records and accuracy would be trivially inflated by majority-class bias. The primary metrics are Macro-AUC and Macro-F1.

3.2. The Morphological Alignment Score

We use two variants of the MAS across the three tasks, because the second improvement introduces class-specific masks that are not available in the simpler binary form.

MAS$_{\text{eval}}$ (Tasks 3 and 4) measures how much of a Grad-CAM heatmap $H \in [0,1]^T$ falls within a binary morphological mask $M \in \{0,1\}^T$ built by NeuroKit2's `ecg_delineate` function (Makowski et al., 2021; Pan & Tompkins, 1985) using Pan-Tompkins R-peak detection on Lead II:

$$\text{MAS}_{\text{eval}} = \frac{\sum_t H_t \cdot M_t}{\sum_t H_t + \varepsilon}. \tag{1}$$

A value near 1.0 indicates the heatmap is concentrated inside P-wave, QRS, and T-wave windows; values near zero indicate the model is attending to isoelectric baseline regardless of the predicted class.

MAS$_{\text{physio}}$ (Task 5) replaces the binary NeuroKit2 mask with class-conditioned Gaussian soft priors derived from class-specific wave weights, averaged over the full test population. This is necessary because the alignment supervision framework introduces per-class physiological targets that the binary mask cannot distinguish. The two metrics are therefore not directly comparable; Section 3.5 defines $\text{MAS}_{\text{physio}}$ precisely.

3.3. Task 3: Baseline Pipeline

Preprocessing. Three leads are used in the baseline: Lead I (index 0), Lead II (index 1), and V1 (index 6). Each lead is independently filtered with a third-order zero-phase Butterworth bandpass at 0.5–40 Hz and then Z-score normalised per recording. The three-lead design is intentional: Lead II provides the clearest P-wave axis, Lead I captures lateral ST changes, and V1 gives access to RBBB/LBBB morphology and posterior MI patterns.

Architecture. The backbone is an 18-layer-style 1D-ResNet. A stem convolution (kernel size 15, stride 2) projects the input from 3 channels to 32 and halves the temporal dimension to 2500. Four progressive stages of residual blocks follow:

- Stage 1: $2\times$ ResBlock$(32 \to 32, k = 15) \to (B, 32, 2500)$
- Stage 2: $2\times$ ResBlock$(32 \to 64, k = 15, s = 2) \to (B, 64, 1250)$
- Stage 3: $2\times$ ResBlock$(64 \to 128, k = 7, s = 2) \to (B, 128, 625)$
- Stage 4: $2\times$ ResBlock$(128 \to 256, k = 7, s = 2) \to (B, 256, 313)$

Stages 1 and 2 use k=15 (30 ms at 500 Hz) to capture steep P-wave and R-peak slopes at high resolution; stages 3 and 4 use k=7 where the effective receptive field already spans ≈224 ms, enough for rhythm context. Global average pooling and a two-layer MLP with sigmoid output produce the five-class multi-label prediction.

Training. Binary cross-entropy with per-class positive weight $w_c = N_c^-/N_c^+$ is used to counteract class imbalance. AdamW (lr= 10^{-3}, wd= 10^{-4}) with ReduceLROnPlateau (patience 5, factor 0.1, monitor val Macro-AUC) trains for 70 epochs; best checkpoint is selected by validation Macro-AUC.

Grad-CAM and MAS_{eval}. Grad-CAM is applied to the last convolutional layer of stage 4. The algorithm back-propagates the logit for class c, computes importance weights $\alpha_k^c = \frac{1}{T}\sum_t \frac{\partial y^c}{\partial A_t^k}$, and forms the heatmap $L^c = \mathrm{ReLU}(\sum_k \alpha_k^c A^k)$. The heatmap is linearly upsampled from 313 to 5000 samples and normalised to $[0, 1]$ before MAS_{eval} is computed against the NeuroKit2 mask on Lead II using Eq. 1.

3.4. Task 4: Preprocessing Ablation and LBWA

LBWA design. We designed Lead-wise Beat-Warp Augmentation to address the shortcoming identified in the SoA survey: standard augmentations either corrupt beat morphology (Gaussian noise) or warp the entire recording, which simultaneously and incorrectly stretches the PR interval, QRS duration, and QT interval. The design draws on two sources: the STAR principle of Nemati (Nemati, 2025), which restricts augmentation to individual RR segments, and the empirical validation by Iwana and Uchida (Iwana & Uchida, 2021) that piecewise warping preserves class-discriminative structure in time-series data.

For each validated RR segment i, LBWA samples a warp factor $c_i \sim \mathcal{U}(1-\alpha, 1+\alpha)$ with α=0.12 and an amplitude factor $a_i \sim \mathcal{N}(1.0, 0.05)$. The segment is resampled to the warped length and back, preserving global signal length:

$$x_i'(t) = a_i \cdot c_i \cdot x_i\left(\frac{t}{c_i}\right), \quad t \in [R_i, R_{i+1}]. \tag{2}$$

R-peaks are identified by a multi-lead majority vote (Pan-Tompkins on each of the three leads via NeuroKit2; peaks confirmed by at least two leads are retained). Any augmented beat that fails a morphological fidelity check (P-QRS-T landmark count and ordering, up to 8 retries) is rejected and resampled. LBWA is applied only to minority classes (MI, STTC, CD, HYP).

SWT denoising. The SWT pipeline decomposes each lead with a Daubechies-8 wavelet at J=8 levels:

$$x(t) = a_J(t) + \sum_{j=1}^{J} d_j(t). \tag{3}$$

The approximation a_8 (baseline wander < 0.5 Hz) is zeroed; the detail d_1 (50 Hz powerline band) is soft-thresholded with a SURE threshold $\lambda = \hat{\sigma}\sqrt{2\log N}$ where $\hat{\sigma} = \mathrm{median}(|d_1|)/0.6745$. The SWT is shift-invariant, avoiding the R-peak jitter introduced by sub-sampling in the standard DWT (Addison, 2005).

Morphological fidelity. Table 2 reports the fidelity check on 50 samples (10 per class). The SWT satisfies the R-peak shift threshold (<1 sample) and R-peak sharpness retention (>90%) but does not achieve the 95% P-wave amplitude retention target, nor does Butterworth. This already hints that neither filter fully preserves the low-amplitude P-wave envelope — a result confirmed by the ablation.

Ablation conditions. The same 1D-ResNet architecture and training procedure from Task 3 is held fixed. Five preprocessing conditions vary only the denoising and augmentation: **A** – Butterworth, no augmentation; **B** – Butterworth + LBWA; **C** – SWT, no augmentation; **D** – SWT + LBWA; **E** – raw ECG with z-score normalisation only (no filter,

Table 2. Morphological fidelity: Butterworth vs SWT denoising. Thresholds: R-shift <1 sample; P-retention >95%; Sharpness >90%.

	Butterworth			SWT		
Class	Shift	P-ret	Sharp	Shift	P-ret	Sharp
NORM	0.08	54.3	62.7	0.03	52.6	96.4
MI	0.18	61.8	60.4	0.06	58.2	96.2
STTC	0.10	68.3	77.3	0.00	67.7	96.5
CD	0.17	63.9	59.9	0.03	61.7	98.1
HYP	0.10	65.8	67.6	0.03	66.5	98.2
Mean	0.13	62.8	65.6	0.03	61.3	97.1

no augmentation). Condition E serves as the lower-bound sanity check in the original design but ended up winning on all metrics.

3.5. Task 5: Physiological Alignment Supervision

Given that raw ECG with z-score normalisation produced the best MAS_{eval}, Task 5 keeps that preprocessing and focuses entirely on the training objective. The core question is whether explicitly regularising the model's class activation maps toward physiologically meaningful ECG regions improves alignment without meaningfully sacrificing classification performance.

Class activation maps. The backbone's `forward` pass is modified to expose the pre-GAP feature map $F \in \mathbb{R}^{B\times C\times T'}$. Class-specific CAMs are computed inline during training as:

$$\text{CAM}[b,k,t] = \text{ReLU}\left(\sum_c W_k^c \cdot F_{b,c,t}\right), \qquad (4)$$

where $W \in \mathbb{R}^{K\times C}$ is the classifier weight matrix. Each CAM is normalised to a probability distribution over time. This is far cheaper than Grad-CAM during training and produces the same class-specific spatial maps.

Class-conditioned physiological priors. Each wave region (P, QRS, ST, T) is represented as a Gaussian soft mask centred at a fixed offset from the detected R-peak:

$$g(t;\mu,\sigma) = \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right). \qquad (5)$$

Offsets and widths encode typical inter-beat timing at 500 Hz (P: -160 ms, σ=60 ms; QRS: 0 ms, σ=40 ms; ST: $+120$ ms, σ=60 ms; T: $+280$ ms, σ=100 ms). The combined prior for class k is a weighted sum of wave masks, downsampled to T' and normalised:

$$\pi_k(t) \propto \sum_{w\in\{P,\,\text{QRS},\,\text{ST},\,T\}} \rho_k^w \cdot g(t;\mu_w,\sigma_w), \qquad (6)$$

where ρ_k^w are class-wave weights encoding clinical knowledge. For example, CD (QRS widening) uses $\rho_{\text{CD}}^{\text{QRS}}$=1.0 and zero elsewhere; STTC uses $\rho_{\text{STTC}}^{\text{ST}}$=0.5, ρ_{STTC}^{T}=0.5.

$\text{MAS}_{\text{physio}}$. For class k, the class-specific MAS is:

$$\text{MAS}_{\text{physio},k} = \mathbb{E}[\text{CAM}_k \cdot \pi_k], \qquad (7)$$

averaged over the test population. Because π_k is a narrow Gaussian-weighted distribution, absolute values are small; the relevant comparison is between ablation conditions.

Four-term alignment loss. Let CAM_{act} denote the stacked CAMs for the active (label= 1) classes in a batch, and π_{act} their corresponding priors. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_a \cdot \text{conf} \cdot \mathcal{L}_{\text{align}} + \lambda_o \cdot \text{conf} \cdot \mathcal{L}_{\text{out}} + \lambda_e \cdot \mathcal{L}_{\text{entropy}}, \qquad (8)$$

where $\mathcal{L}_{\text{cls}}$ is binary cross-entropy with per-class positive weights, and conf is the delineation confidence score returned during mask pre-computation (poor-quality masks produce weak rather than incorrect supervision).

$\mathcal{L}_{\text{align}}$ is an asymmetric KL divergence:

$$\mathcal{L}_{\text{align}} = \alpha\, \text{KL}(\text{CAM} \,\|\, \pi) + (1-\alpha)\, \text{KL}(\pi \,\|\, \text{CAM}), \qquad (9)$$

with α=0.7. The mode-seeking term $\text{KL}(\text{CAM} \,\|\, \pi)$ carries more weight because pure mean-seeking KL produces negligible gradient when the CAM is initially diffuse, which is exactly when the alignment signal is most needed.

$\mathcal{L}_{\text{out}}$ penalises CAM mass outside the physiological region:

$$\mathcal{L}_{\text{out}} = \mathbb{E}[\text{CAM} \cdot \mathbf{1}_{\text{outside}}], \qquad (10)$$

where "outside" is defined as the bottom $(1-q)$ timesteps by prior probability with q=0.35.

$\mathcal{L}_{\text{entropy}} = -\mathbb{E}[\text{CAM} \cdot \log \text{CAM}]$ sharpens temporal localisation by minimising entropy.

A linear warmup keeps all λ values at zero for the first few epochs while early CAMs stabilise. Only active classes contribute to alignment losses; negative labels have no physiological target to align toward.

Ablation conditions. Four conditions are trained on the same raw + z-score data and the same model: **Baseline** ($\mathcal{L}_{\text{cls}}$ only), **Align** ($+\mathcal{L}_{\text{align}}$), **Align+Out** ($+\mathcal{L}_{\text{align}} + \mathcal{L}_{\text{out}}$), **Full** (all four terms).

4. Results

4.1. Task 3: Baseline Performance

The baseline 1D-ResNet trained on Butterworth-filtered three-lead inputs converges steadily over 70 epochs. Grad-CAM maps on the test set show partial localisation to the QRS complex across most superclasses, with weaker, less consistent localisation to P-wave and ST-segment regions for CD and STTC respectively. $\mathrm{MAS}_{\mathrm{eval}}$ for the Butterworth no-augmentation condition (Condition A from Task 4, which replicates the baseline preprocessing) is reported in Table 3.

4.2. Task 4: Preprocessing Ablation

Table 3 shows the five-condition results. Several findings are worth noting directly.

First, raw ECG beats every filtered variant on all three metrics. The gap is most pronounced in $\mathrm{MAS}_{\mathrm{eval}}$: 0.711 for raw versus 0.591 for SWT-only and 0.478 for Butterworth-only. This is not a small margin.

Second, LBWA hurt performance in both filtering regimes. Butterworth+LBWA dropped $\mathrm{MAS}_{\mathrm{eval}}$ from 0.478 to 0.456 compared to Butterworth alone; SWT+LBWA dropped from 0.620 to 0.535. LBWA also reduced Macro-F1 in both cases. Given this consistent negative result, LBWA is dropped and raw+z-score is carried forward.

Third, the per-class breakdown reveals that MI and HYP benefit the most from raw preprocessing in terms of AUC (0.909 and 0.818 respectively for Condition E), while CD benefits most in $\mathrm{MAS}_{\mathrm{eval}}$ under raw signals. Figures 1 and 2 show the Grad-CAM overlays for Conditions A and E respectively; the improvement in heatmap-mask overlap is visible across MI, STTC, and CD.

4.3. Task 5: Physiological Alignment Supervision

Table 5 reports absolute metrics and Table 6 shows deltas relative to the Baseline condition. Note that $\mathrm{MAS}_{\mathrm{physio}}$ values are not comparable to the $\mathrm{MAS}_{\mathrm{eval}}$ values in Task 4: they use different masks (Gaussian priors versus binary NeuroKit2) and are averaged over the full test population rather than representative samples.

The alignment supervision delivers a 58% relative increase in mean class-specific MAS (0.00451 to 0.00713) with an AUC gain of $+0.0008$. The CD class shows the largest absolute gain: $\mathrm{MAS}_{\mathrm{physio,CD}}$ goes from 0.001 to 0.011 under the Full condition, roughly a tenfold increase. The effect on HYP is also positive (0.009 to 0.009 under Align, essentially unchanged). Classification metrics are well-preserved: the Align condition actually gives the best Macro-F1 (0.6729 vs 0.6690 for Baseline).

5. Discussion

5.1. Why Raw Signal Outperforms Filtered Input

The most counterintuitive result in this work is that removing all filtering improves both classification and morphological alignment. The explanation is not complicated once you look at the fidelity numbers in Table 2. Both Butterworth and SWT retain only 62–68% of P-wave amplitude on average. The P-wave sits predominantly in the 0.5–5 Hz band; a Butterworth cutoff at 0.5 Hz clips that band's lower edge, smoothing the wave. The SWT is better for R-peak sharpness (97% retention vs 66%), but P-wave retention is nearly identical to Butterworth at 61–68%. A model trained on pre-smoothed P-waves cannot be expected to learn P-wave-dependent patterns reliably, and the Grad-CAM maps confirm this: conditions A and C both show weak localisation to P-wave regions for CD, the class most sensitive to P-wave timing.

The raw signal presumably helps because the ResNet can learn to ignore low-frequency drift (which appears consistently across training samples) while retaining the full amplitude information of every wave. Z-score normalisation removes the inter-recording amplitude scale but leaves intra-recording morphological shape intact.

5.2. Why LBWA Did Not Help

We expected LBWA to improve both generalisation and alignment by exposing the model to more authentic morphological variation while preserving the P-QRS-T ordering. The opposite happened. There are a few plausible explanations worth considering.

Piecewise time-warping within validated RR segments may still introduce subtle distortions at segment boundaries that the morphological fidelity check does not catch. The model could be learning to average over these boundary artefacts rather than sharpening its representation of the beat interior. Alternatively, LBWA's amplitude perturbation ($a_i \sim \mathcal{N}(1.0, 0.05)$) at 5% variation may be enough to confuse the Grad-CAM calibration used by $\mathrm{MAS}_{\mathrm{eval}}$, even if the underlying model has not changed substantially. Either way, the negative result is clear and consistent across both filtering regimes. We would have needed to train on substantially more epochs with LBWA or tune α further to say anything more definitive about whether the failure is fundamental.

5.3. Alignment Supervision: What Works and What Does Not

The CD result (tenfold $\mathrm{MAS}_{\mathrm{physio}}$ increase) is the clearest success story. QRS widening is a spatially compact, reproducible target: the Gaussian prior for CD centres on t_R with $\sigma = 40$ ms, which maps to a narrow 40-sample window at 500 Hz. When the alignment loss pushes the CAM toward that window, there is little ambiguity about where the region is, so the gradient signal is clean. STTC and MI see smaller gains (per Table 6), likely because their ST and T-wave priors overlap spatially and the multi-label nature of PTB-XL means both labels are often active simultaneously.

The low absolute values of $\mathrm{MAS}_{\mathrm{physio}}$ (0.007 at best) reflect

Table 3. Five-condition preprocessing ablation (Task 4). $\mathrm{MAS}_{\mathrm{eval}}$ computed via Grad-CAM vs. NeuroKit2 binary P/QRS/T mask. Best result per column in bold.

Cond.	Preprocessing	Macro-AUC	Macro-F1	$\mathrm{MAS}_{\mathrm{eval}}$	$\mathrm{AUC}_{\mathrm{N}}$	$\mathrm{AUC}_{\mathrm{MI}}$	$\mathrm{AUC}_{\mathrm{STTC}}$	$\mathrm{AUC}_{\mathrm{CD}}$	$\mathrm{AUC}_{\mathrm{HYP}}$
A	Butterworth, no aug	0.8975	**0.6952**	0.478	0.929	0.906	0.915	0.924	0.814
B	Butterworth + LBWA	0.8946	0.6722	0.456	0.931	0.896	0.911	0.921	0.814
C	SWT, no aug	0.8984	0.6840	0.620	0.931	**0.912**	0.913	**0.927**	0.809
D	SWT + LBWA	0.8931	0.6582	0.535	0.924	0.901	0.914	0.915	0.812
E	**Raw + Z-score, no aug**	**0.8987**	0.6949	**0.711**	0.930	0.909	**0.916**	0.919	**0.818**

Table 4. Per-class $\mathrm{MAS}_{\mathrm{eval}}$ breakdown by condition.

Cond.	NORM	MI	STTC	CD	HYP
A	0.261	0.395	0.255	0.621	0.330
B	0.281	0.770	0.604	0.392	0.812
C	0.577	0.319	0.857	0.529	0.673
D	0.492	0.290	0.555	0.571	0.347
E	**0.656**	**0.839**	**0.775**	**0.762**	**0.525**

Table 5. Task 5 alignment ablation. $\mathrm{MAS}_{\mathrm{physio}}$ is the mean class-specific MAS averaged over the test population. Epochs = 30.

Ablation	Macro-AUC	Macro-F1	$\mathrm{MAS}_{\mathrm{physio}}$	Std
Baseline	0.8960	0.6690	0.00451	0.169
Align	0.8964	0.6729	0.00712	0.167
Align+Out	0.8963	0.6727	0.00713	0.152
Full	**0.8968**	0.6716	**0.00713**	0.152

the narrowness of the Gaussian priors rather than failure of alignment. A model attending perfectly to the QRS window would still produce a low $\mathrm{MAS}_{\mathrm{physio}}$ if the prior is narrow relative to the signal length. The informative quantity is the relative improvement, not the absolute score.

Macro-F1 tells a slightly nuanced story: Align has the best F1 (0.6729), while Full (which adds entropy sharpening) has slightly lower F1 (0.6716) but the best AUC (0.8968). The entropy loss sharpens localisation at the cost of some probability calibration, suggesting a tradeoff between spatial precision and class separation that a more sophisticated schedule might resolve.

Table 6. Task 5 deltas relative to Baseline. CD gains most because QRS widening maps to the narrowest, most precise physiological prior (σ=40 ms).

Ablation	ΔAUC	ΔF1	$\Delta\mathrm{MAS}_{\mathrm{p}}$	ΔCD	ΔSTTC
Align	+.0004	+.0039	+.0026	+.0096	+.0023
Align+Out	+.0003	+.0037	+.0026	+.0094	+.0024
Full	+.0008	+.0026	+.0026	+.0097	+.0023

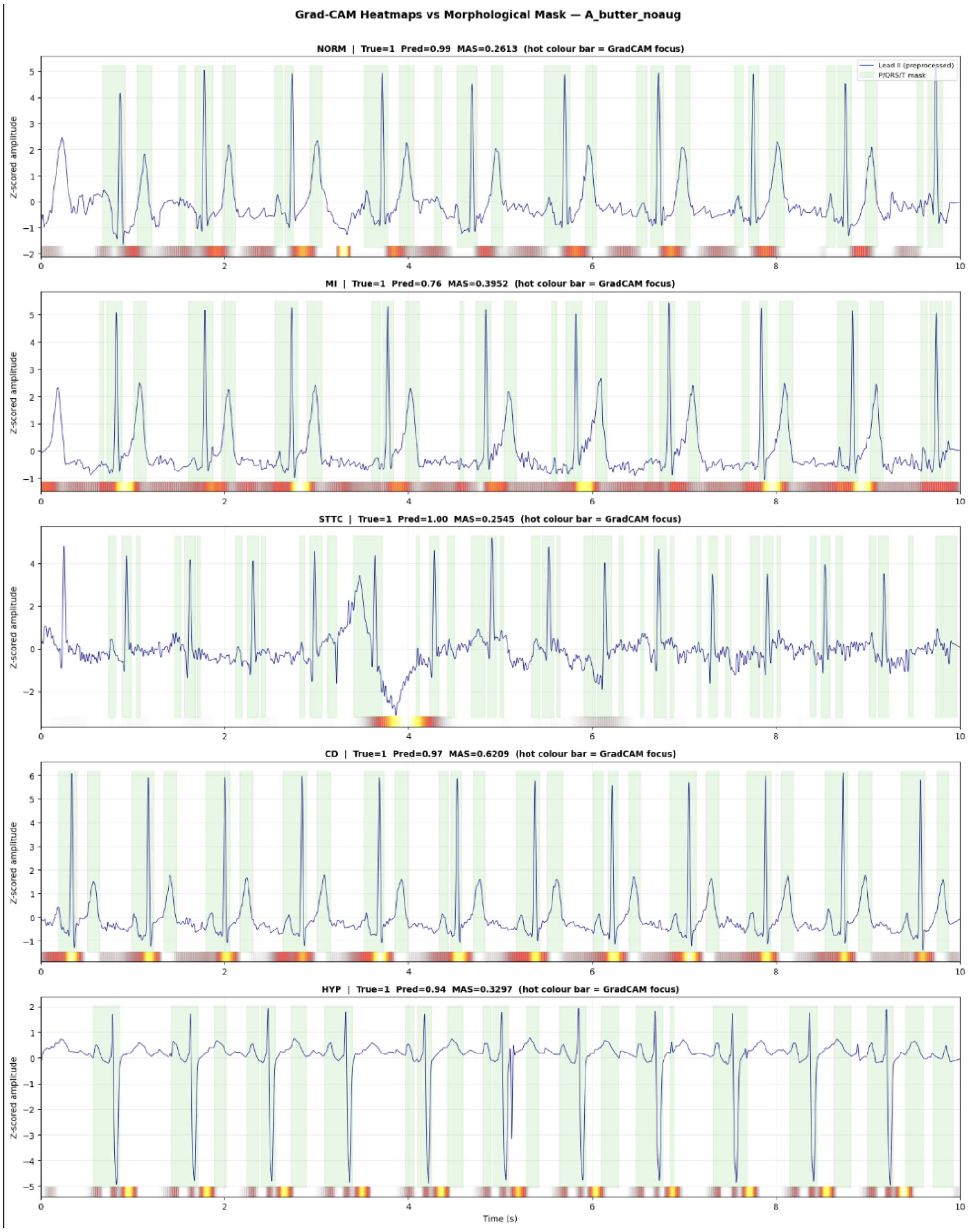

Figure 1. **Condition A (Butterworth filter, no augmentation) — all five superclasses.** Grad-CAM heatmaps overlaid on Lead II. Hot colourbar at the bottom of each panel marks where the model concentrates attention; green shading marks the NeuroKit2-derived P/QRS/T morphological mask. Per-class MAS_{eval}: NORM = 0.261, MI = 0.395, STTC = 0.255, CD = 0.621, HYP = 0.330 (mean = 0.478). The model attends largely to the high-amplitude QRS complex and isoelectric baseline, with weak signal in the ST and P-wave windows for MI and STTC.

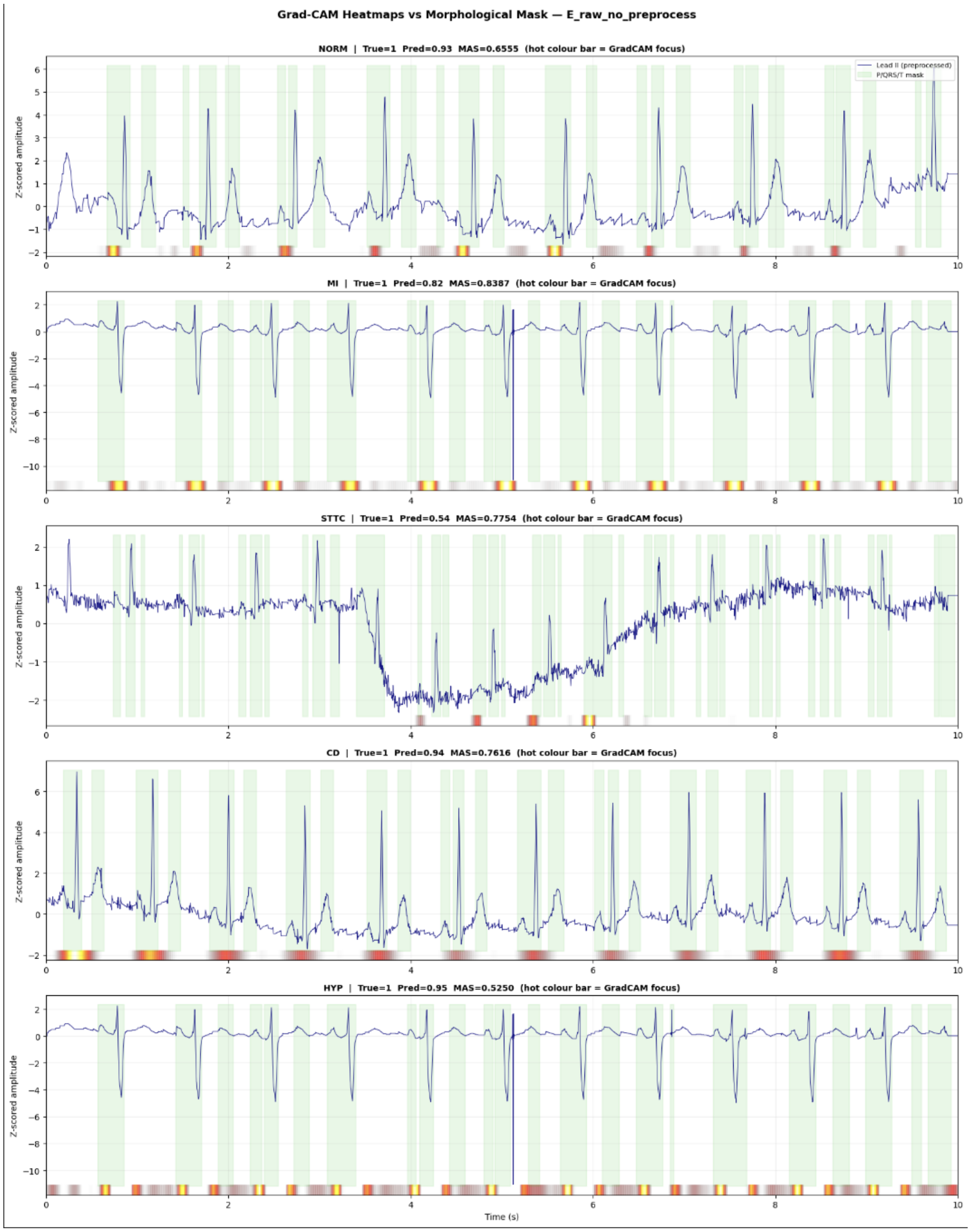

Figure 2. **Condition E (raw + z-score, no filter) — all five superclasses.** Same architecture and training procedure as Condition A; only preprocessing changed. Per-class MAS_{eval}: NORM = 0.656, MI = 0.839, STTC = 0.775, CD = 0.762, HYP = 0.525 (mean = 0.711). The heatmap colourbar aligns visibly with the green mask regions across all classes, consistent with the 48% relative MAS improvement over Condition A.

6. Conclusion

Three things emerged clearly from this work.

Butterworth filtering, widely used in ECG deep learning pipelines, measurably suppresses P-wave amplitude below the threshold needed for reliable P-wave-dependent learning. Raw ECG with z-score normalisation is a better starting point for morphological classification on PTB-XL, and this likely generalises to other clean clinical datasets.

Beat-preserving augmentation designed to simulate physiological heart rate variability (LBWA) did not improve alignment or accuracy in this study. Whether this is a fundamental property of beat-warping or a consequence of our specific implementation and hyperparameter choices remains open.

Explicitly regularising the model's class activation maps toward class-specific physiological priors does improve morphological alignment in a measurable way, with the Conduction Disturbance class showing the clearest gain. The classification cost is negligible.

Several limitations constrain these findings. The experiment is restricted to single-lead input in Tasks 4 and 5, which limits the clinical scope. The physiological priors are defined by fixed offsets from R-peaks, so records with irregular heart rates or pathological conduction delays will receive noisy supervision. The MAS metrics evaluate a single sample per class in Task 4 and narrow Gaussian windows in Task 5; neither fully captures whether the model reasons about morphology the way a cardiologist does.

Future work should pursue learnable mask boundaries that adapt to per-patient conduction velocities, multi-head attention where each head is supervised toward a distinct wave region, and contrastive pretraining to push morphologically distinct classes further apart in feature space. ProtoECGNet (Sethi et al., 2025) defines the ceiling for intrinsic explainability on this dataset; closing the gap between alignment-supervised models and prototype-based reasoning is the most natural next step.

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